DO NOW — RECALL

Lesson 125 · Assembly Language and the Little Man Com

Name: _____ Class: _____ Date: _____

1. What is the purpose of a function return value?

2. After pushing A then B onto a stack, which is popped first?

3. What is the output of a full adder with A=1, B=1 and carry-in=0?

4. Why is the program counter important in a stored-program computer?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It passes a computed result to the caller.

2. Model answer

B, since a stack uses LIFO.

3. Model answer

Sum 0, carry-out 1.

4. Model answer

It tracks the address of the next instruction to fetch.

By the end, you can…

• Explain the roles of the accumulator, program counter and numbered memory mailboxes in the LMC model.

• Use INP, OUT, LDA, STA, ADD, SUB, BRA, BRZ, BRP, HLT and DAT correctly in simple programs.

• Trace a short stored-program sequence and give the contents of relevant memory and registers.

Vocabulary — know these terms

LMC — Educational model of a stored-program accumulator computer.

accumulator — Register holding current working data and arithmetic results.

program counter — Register holding address of the next instruction to fetch.

mnemonic — Symbolic name representing a machine instruction.

NEW LEARNING

Read, annotate and explain

Mailboxes

Instructions and numeric data occupy addressed memory mailboxes. DAT reserves a data location. Programs execute the instructions addressed by the program counter.

Accumulator and PC

INP reads into ACC; LDA reads memory into ACC; STA writes ACC to memory. The PC normally advances after a fetch, unless a branch selects a new address.

Branches and stopping

ADD and SUB update ACC; OUT prints ACC; BRA always branches; BRZ tests for zero; BRP tests a non-negative result in standard LMC; HLT ends the program.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Input is 7. Trace this program, including the accumulator, data mailbox and output.

Track each change

00 INP: ACC = 7.

01 STA 10: mailbox 10 becomes 7; ACC remains 7.

02 LDA 10: ACC = 7.

03 ADD 10: ACC = 14.

04 OUT: output is 14 .

05 HLT: stop. Data mailbox 10 is separate from instruction addresses 00–05.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain what the PC, accumulator and memory mailboxes represent in LMC. [6]

Question 2

Trace INP; STA 30; LDA 30; ADD 31; OUT; HLT, with input 5 and mailbox 31 containing 3. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write LMC instructions to read an integer and output it, using a BRZ branch when zero. Explain how you would test both paths. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In LMC, STA n loads the accumulator with the contents of mailbox n.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.